INFO2604 Information Systems Security

Tutorial - 2

1. Section A: Short Answer Questions

a) Define cryptography and cryptanalysis.
b) What is the difference between plaintext and ciphertext?
c) What is symmetric cryptography?
d) State one advantage and one disadvantage of symmetric cryptography.
e) What is asymmetric cryptography?
f) Distinguish between stream ciphers and block ciphers.
g) What are the two basic building blocks of cryptographic systems?
h) Define substitution cipher.
i) Define transposition cipher.
j) What is a monoalphabetic cipher?

2. Section B: Conceptual Questions

a) What is a polyalphabetic cipher and why is it more secure than monoalphabetic ciphers?
b) Explain diffusion in cryptographic systems.
c) Explain confusion in cryptographic systems.
d) Define a cryptosystem.
e) State the condition for correct decryption in a cryptosystem.

3. Section C: Mathematical Foundations

a) State the Fundamental Theorem of Arithmetic.
b) Define greatest common divisor (GCD).
c) What does it mean for two numbers to be relatively prime?
d) Use the Euclidean Algorithm to find gcd(1230, 450).
e) When do two integers have a multiplicative inverse modulo n?

4. Section D: Classical Cryptosystems

a) Describe the Caesar Cipher.
b) Encrypt “UWI ST AUGUSTINE” using Caesar cipher with k = 7.
c) Why is the Caesar cipher insecure?
d) How many possible keys exist in a substitution cipher?

e) Define the Vigenère cipher.
f) Encrypt “notsecure” using the Vigenère cipher with keyword “dog”.

5. Section E: One-Time Pad (OTP)
a) State the encryption and decryption equations for OTP.
b) List the four rules of a One-Time Pad.
c) Why is OTP considered perfectly secure?
d) State one disadvantage of the One-Time Pad.

6. Section F: Block and Product Ciphers
a) What is a block cipher?
b) Define the avalanche effect.
c) Define completeness in block ciphers.
d) What is a product cipher?
e) State two main components of a product cipher.

7. Section G: DES
a) How many rounds does DES use?
b) What is the block size and key size of DES?
c) What cipher structure does DES use?